

Pt. 2 - Fitting Points to a Parabola / Line

Fitting to a line:

$$y = mx + b \Rightarrow y = c_0 + c_1 x$$

$$\begin{aligned} e(x) &= y^*(x) - y(x) \\ &= c_0 + c_1 x - y(x) \end{aligned}$$

determine c_0, c_1 to minimize $p(c_0, c_1) = \|e(x)\|_2^2$

$$p = (c_0 - 0.5)^2 + (c_0 + 2c_1 - 3.5)^2 + (c_0 + c_1 - 1.5)^2 + (c_0 + 3c_1 - 7.5)^2$$

$$\frac{\partial p}{\partial c_0} = \frac{\partial p}{\partial c_1} = 0$$

$$\frac{\partial p}{\partial c_0} = 2(c_0 - 0.5)(1) + 2(c_0 + 2c_1 - 3.5)(1) + 2(c_0 + c_1 - 1.5)(1) + 2(c_0 + 3c_1 - 7.5)(1)$$

$$0 = 8c_0 + 12c_1 - 26$$

$$\frac{\partial p}{\partial c_1} = 2(c_0 - 0.5)(0.5) + 2(c_0 + 2c_1 - 3.5)(2) + 2(c_0 + c_1 - 1.5)(1) + 2(c_0 + 3c_1 - 7.5)(3)$$

$$0 = 12c_0 + 28c_1 - 62$$

$$\begin{cases} 8c_0 + 12c_1 = 26 \\ 12c_0 + 28c_1 = 62 \end{cases}$$

$$c_1 = 2.3 \quad c_0 = -0.2$$

$$\text{Best line of fit: } y = -0.2 + 2.3x$$

Fitting to a parabola:

$$y = Bx^2 + Cx + D \Rightarrow y = c_0 + c_1 x + c_2 x^2$$

We need to satisfy the following equations for $(0, 0.5), (2, 3.5), (1, 1.5), (3, 7.5)$

$$0.5 = 0x^2 + 0x + D$$

$$3.5 = 2^2 x^2 + 2x + D$$

$$1.5 = 1^2 x^2 + 1x + D$$

$$7.5 = 3^2 x^2 + 3x + D$$

In matrix form:

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 4 & 2 & 1 \\ 1 & 1 & 1 \\ 9 & 3 & 1 \end{pmatrix}$$

$$x = \begin{pmatrix} B \\ C \\ D \end{pmatrix}$$

$$b = \begin{pmatrix} 0.5 \\ 3.5 \\ 1.5 \\ 7.5 \end{pmatrix}$$

$$A^T A = \begin{pmatrix} 0 & 4 & 1 & 9 \\ 0 & 2 & 1 & 3 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 4 & 2 & 1 \\ 1 & 1 & 1 \\ 9 & 3 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 98 & 36 & 14 \\ 36 & 14 & 6 \\ 14 & 6 & 4 \end{pmatrix}$$

$$A^T b = \begin{pmatrix} 0 & 4 & 1 & 9 \\ 0 & 2 & 1 & 3 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0.5 \\ 3.5 \\ 1.5 \\ 7.5 \end{pmatrix}$$

$$= \begin{pmatrix} 83 \\ 31 \\ 13 \end{pmatrix}$$

$$\left(\begin{array}{ccc|c} 98 & 36 & 14 & 83 \\ 36 & 14 & 6 & 31 \\ 14 & 6 & 4 & 13 \end{array} \right) \xrightarrow{\text{RREF using calculator}} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 3/4 \\ 0 & 1 & 0 & 1/20 \\ 0 & 0 & 1 & 11/20 \end{array} \right)$$

$$\hat{x} = \begin{pmatrix} 3/4 \\ 1/20 \\ 11/20 \end{pmatrix}$$

Best-fit parabola: $y = 3/4 x^2 + 1/20 x + 11/20$